Docker and Containers

Docker is a platform used to package and run applications in containers. A container provides an isolated environment in which an application and many of its dependencies can be bundled together. The goal is to make software behave more consistently across different machines and environments.

A Docker image is a packaged template used to create containers. Images are commonly built from a Dockerfile, which contains instructions describing how the image should be constructed. A Dockerfile can specify a base image, copy application files, install dependencies, expose a port, and define the command that runs when a container starts.

A container is a running instance of an image. Multiple containers can be created from the same image. Containers are generally lighter than traditional virtual machines because they share the host operating system kernel rather than requiring a complete guest operating system for every application.

Docker images can be stored in registries. Docker Hub is a popular public registry, while organizations can operate private registries. Developers can pull images from a registry, run them locally, and push updated images. Images are commonly identified using repository names and tags such as version numbers.

Docker Compose provides a way to define and run applications consisting of multiple containers. A Compose file can describe services such as a web application, database, and cache. Large organizations may use Kubernetes to schedule containers across machines, restart failed workloads, scale services, and manage networking.